

Adaptive Capillary Skin (ACS)

Open falsifiable design of a biomimetic fluidic material integrating atmospheric water handling, hygromorphic regulation, evaporative cooling, electrokinetic self-sensing and damage-triggered hydraulic isolation

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Version 0.1 | 19 August 2026

COMPUTATIONAL DESIGN STUDY - NO PHYSICAL EXPERIMENTAL DATA

All numerical results in this document are generated by explicit computational models and assumptions. They must not be represented as measurements, proof of physical feasibility, or clinical/environmental performance. The document is designed to make the concept falsifiable before fabrication.

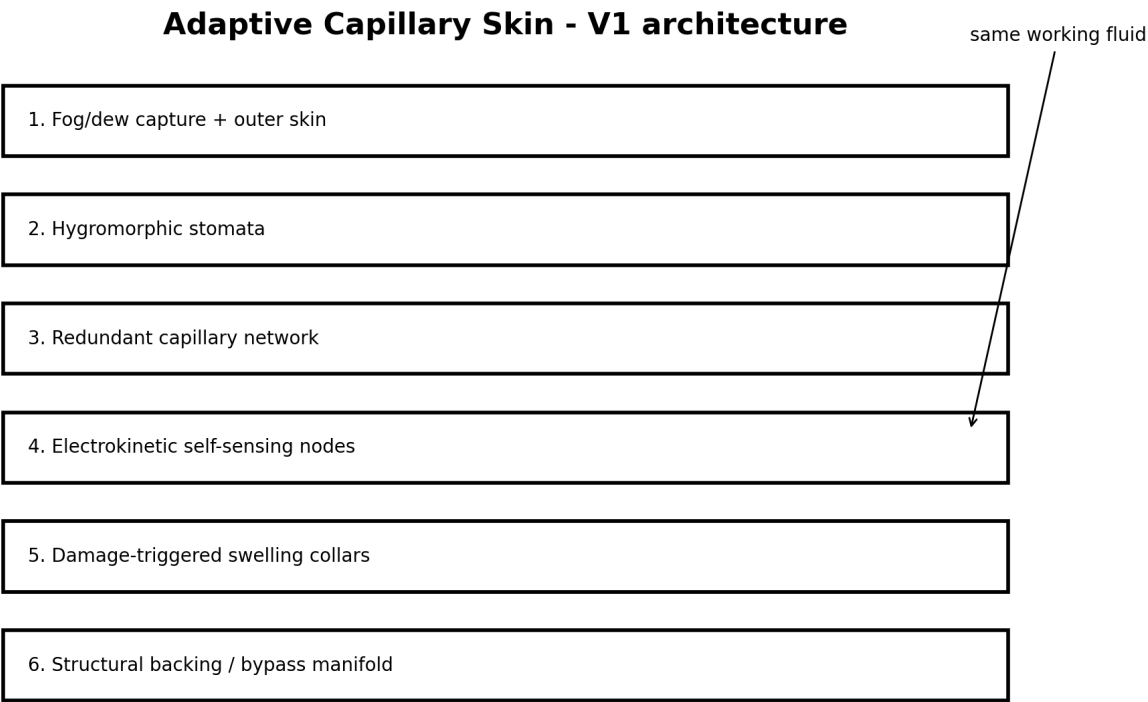
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Executive summary

Adaptive Capillary Skin (ACS) is a proposed material architecture in which a distributed capillary network uses one working fluid - water - as a transport medium, thermal medium, state carrier and damage-response trigger. The design deliberately separates itself from the idea of a single “smart sensor” or a multi-component machine controlled by a central processor. Instead, it asks whether the physical state of the material can perform part of the control logic.

The central falsifiable hypothesis is that a puncture or local rupture in a redundant capillary network can create a pressure/flow perturbation that automatically redirects working fluid toward a swelling collar surrounding the damaged branch. If the collar reaches an appropriate swelling ratio quickly enough, it should reduce local leak conductance while alternate channels preserve useful flow. An electrokinetic or hydrovoltaic readout may simultaneously report the altered flow state. The present work tests only the hydraulic logic computationally and defines experimental tests for the remaining functions.

The model is intentionally capable of failure. Under broad parameter distributions, only 27.3% of 1,000 Monte Carlo runs met all preregistered gates. After converting the failure modes into a design envelope - principally constraining hydrogel response time and residual leak conductance - 99.7% of 1,000 virtual runs met the same gates. These numbers are conditional model results, not empirical validation.



Core hypothesis: damage creates the hydraulic perturbation that drives its own local isolation.

Figure 1. ACS V1 conceptual architecture. The same working fluid is proposed to couple water handling, passive actuation, state sensing and damage response.

1. Origin of the design and biological abstraction

ACS is not an attempt to copy a single organism. It combines abstracted physical strategies observed across biological and bioinspired systems: hygromorphic plant tissues that move without motors, surfaces that manipulate droplets through wettability contrasts, vascular networks that distribute fluid redundantly, and damage responses in which local physical changes alter transport. The engineering objective is to reproduce the logic, not the appearance, of these systems.

Table 1. Biological abstractions used in ACS.

Biological / physical inspiration	Abstracted principle	ACS interpretation
Stomata and hygromorphic plant tissues	Environmental moisture alters local geometry or permeability	Humidity-responsive gates modulate access to the vascular layer without powered actuators
Pine-cone hygromorphs	Differential swelling converts humidity into bending	Passive aperture elements with a designed RH hysteresis
Fog-harvesting surfaces	Wettability and geometry promote nucleation, coalescence and drainage	Outer capture layer directs fog/dew/condensate toward capillary inlets

Biological / physical inspiration	Abstracted principle	ACS interpretation
Plant/fungal vascular networks	Distributed transport with alternate paths	Redundant microchannel mesh preserves function after local blockage or rupture
Biological wound response	Damage changes local transport and recruits repair material	A leak increases local wetting/flow and triggers swelling of a constricting collar

2. Novelty boundary: what is and is not claimed

The literature already contains passive hygromorphic actuators, fog/dew collectors, hydrovoltaic generators, microvascular materials, self-healing systems and passive radiative/evaporative cooling concepts. Therefore none of those components is presented here as independently new. A preliminary literature and patent screen also shows extensive prior art around self-healing microvascular networks and moisture-driven electrical generation. A professional patentability or freedom-to-operate search would still be required before any legal conclusion.

Candidate architectural contribution. The specific hypothesis tested here is the integration of those physical functions around one distributed working-fluid network, with damage-induced hydraulic redistribution acting as the local control signal for passive isolation, while the network remains available for water transport, heat transfer and state sensing. This document should therefore be read as an open, testable design disclosure rather than a declaration that patent-level novelty has been established.

3. ACS V1 architecture

Table 2. Proposed first-build architecture. Dimensions and materials are design targets, not validated specifications.

Layer / module	Proposed V1 implementation	Primary function	Falsifiable output
1. Capture skin	Patterned hydrophilic/hydrophobic or textured upper layer	Fog/dew/condensate collection and drainage	Collected mass per area per time
2. Hygromorphic gate	Cellulose-filled bilayer or equivalent humidity-responsive flap	Passive opening/closing of micro-inlets	Aperture vs RH, hysteresis, response time
3. Capillary vascular layer	Redundant 2-D network, target hydraulic diameter about 150-250 μm for first fabrication	Pump-free local redistribution	Flow map, priming pressure, useful throughput
4. State-sensing node	Charged porous insert and electrode pair, or other electrokinetic readout	Detect local pressure/flow change	Voltage or current effect size vs noise and controls
5. Damage-isolation collar	Confined swellable hydrogel around vulnerable branch / port	Convert damage wetting into reduced leak conductance	Leak reduction and isolation time

Layer / module	Proposed V1 implementation	Primary function	Falsifiable output
6. Backing and bypass manifold	Impermeable structural support with alternate paths	Maintain integrity and reroute flow	Fraction of useful flow retained after damage

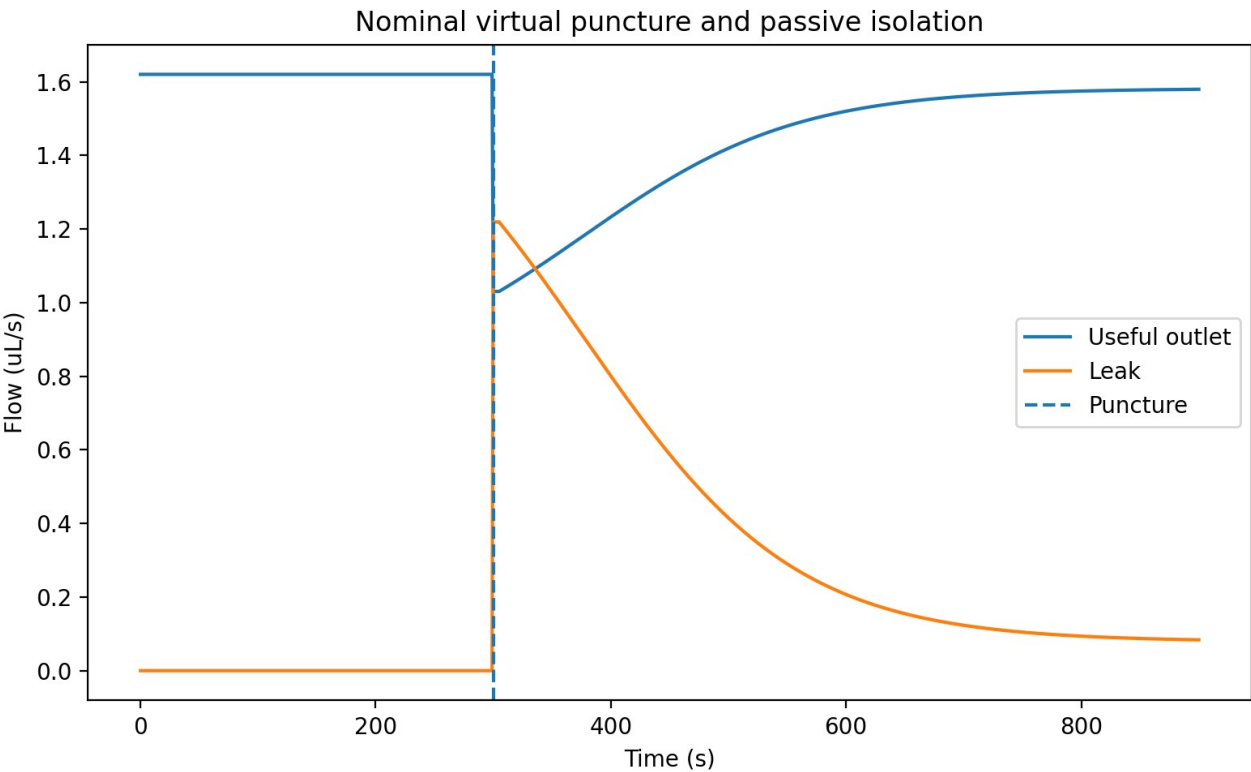


Figure 2. Nominal virtual puncture test. A leak is introduced at $t = 300$ s. The modeled damage collar progressively reduces leak conductance while redundant paths preserve outlet flow.

4. Governing model

The computational model is deliberately low-order. It is intended to answer a design question - whether the proposed hydraulic feedback logic can be internally consistent under plausible parameter variation - rather than to predict a final manufactured device. Every equation below identifies a future calibration target.

4.1 Channel hydraulic conductance

Each channel is approximated as a circular laminar conduit. Its conductance G is calculated from the Hagen-Poiseuille relation:

$$G = \pi r^4 / (8 \mu L) \quad (1)$$

where r is channel radius, μ is dynamic viscosity and L is channel length. Flow on an edge i - j is $Q_{ij} = G_{ij} (P_i - P_j)$. At internal nodes, mass conservation is enforced by requiring the algebraic sum of flows and any leak term to be zero. The fourth-power radius dependence makes geometry variation an intentionally strong stress test.

4.2 Capillary driving pressure

Capillary priming is not dynamically simulated in V0.1; instead, boundary pressure is prescribed. The physical build must determine whether the geometry and surface chemistry can supply the required pressure. The relevant scale is the Young-Laplace relation:

$$\Delta P_{cap} = 2 \gamma \cos(\theta) / r \quad (2)$$

For water, favorable wetting and radii on the order of 10^{-4} m can produce pressures on the order of hundreds to thousands of pascals. This makes the nominal 1 kPa source pressure a useful design-scale stress test, not a claim that a fabricated panel will automatically achieve it.

4.3 Damage and passive isolation law

A puncture is represented as a leak conductance G_l at the central node. After a 5 s wetting delay, the collar response is represented by an exponential decay toward a residual leak conductance:

$$G_l(t) = G_{min} + (G_0 - G_{min}) \exp[-(t - t_d - t_{delay})/\tau] \quad (3)$$

The key design parameters are therefore the hydrogel response time constant τ and the residual conductance ratio G_{min}/G_0 . The model does not assume a specific polymer chemistry; physical testing must determine whether a candidate collar can achieve the required combination of swelling speed, confinement force, reversibility and fatigue resistance.

4.4 Idealized electrokinetic self-sensing surrogate

To test whether the hydraulic perturbation could in principle be visible electrically, a Smoluchowski-type scaling is used as a surrogate:

$$\Delta V = (\epsilon \zeta / (\mu \sigma)) \Delta P \quad (4)$$

where ϵ is fluid permittivity, ζ is zeta potential, σ is conductivity and ΔP is the local pressure difference. Real hydrovoltaic and streaming-potential devices depend on pore size, surface charge, electrodes, ionic concentration, impedance and electrochemical effects. Consequently the simulated sensor effect is not an expected device voltage; it is a sensitivity screen that identifies whether the hydraulic signal is large enough to justify building a readout experiment.

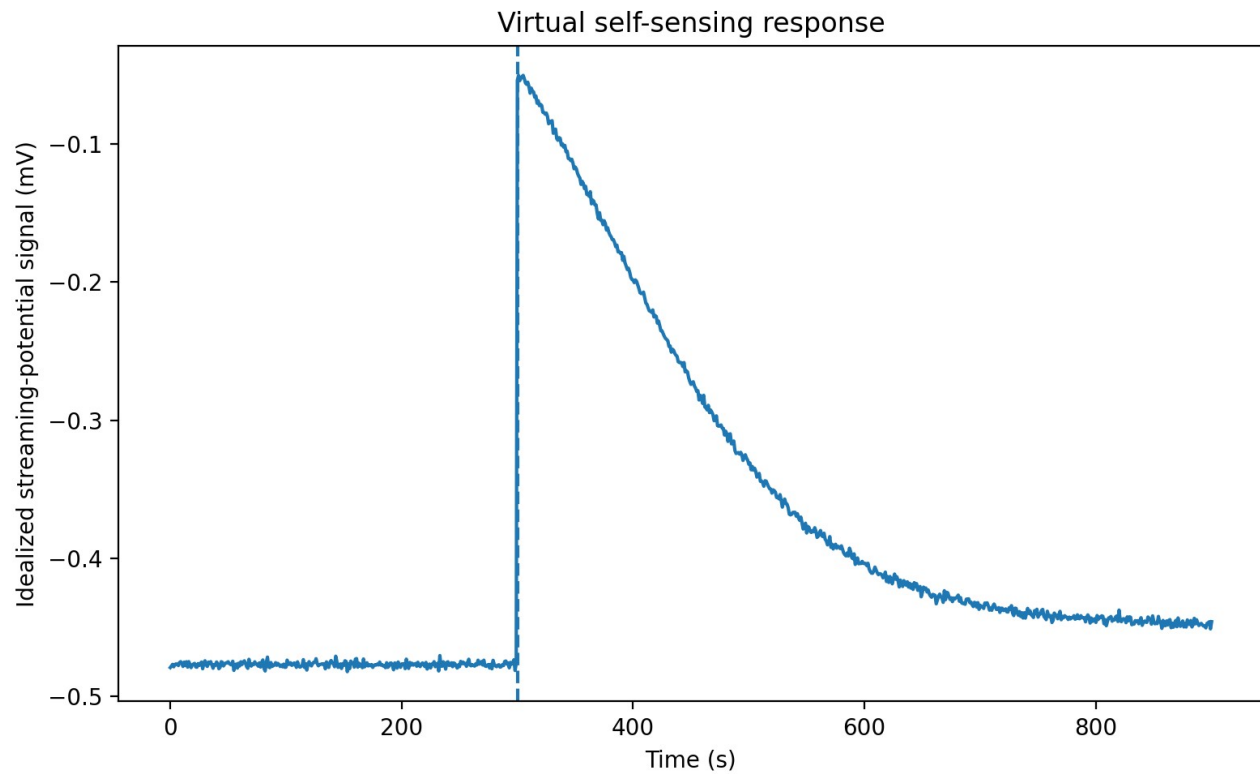


Figure 3. Idealized electrokinetic state signal in the nominal simulation. The plotted voltage is generated by a simplified pressure-to-potential surrogate and assumed readout noise; it is not an experimentally measured hydrovoltaic output.

4.5 Hygromorphic gate model

The opening and closing branches of the humidity-responsive gate are represented by logistic curves with different centers, producing a simple hysteresis loop. The chosen 65% RH opening center and 55% RH closing center are design placeholders. The physical gate experiment must measure actual curvature/aperture, hysteresis, response time and fatigue under repeated humidity cycles.

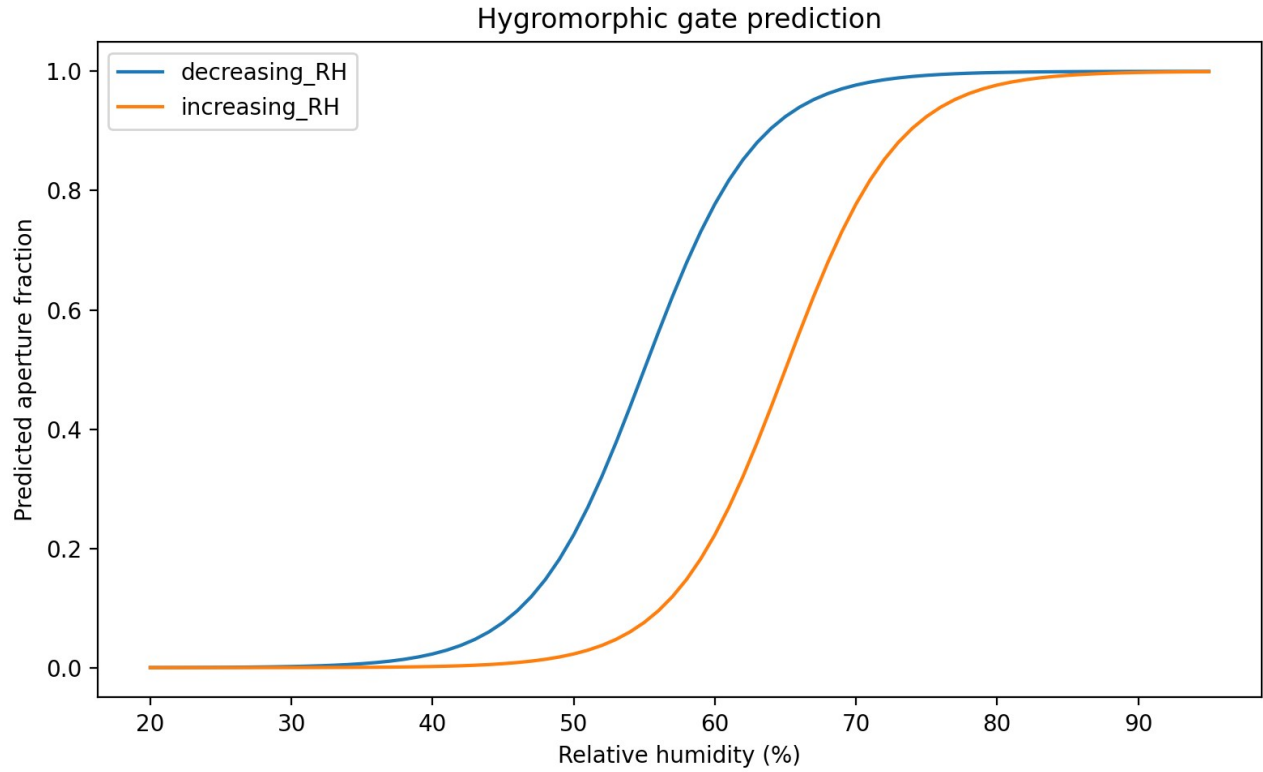


Figure 4. Proposed hygromorphic gate response used for design visualization. This is a parametric prediction, not empirical material data.

4.6 Evaporative cooling and atmospheric-water bounds

Two separate bounding calculations are supplied so that the project does not overstate water or cooling performance. Fog collection is estimated from liquid water content, air velocity, panel area and an assumed collection efficiency. Evaporative cooling is bounded by the latent heat associated with an assumed evaporation rate and by the psychrometric wet-bulb limit. Neither model includes detailed mass-transfer coefficients, solar load, radiative exchange, wind profiles, salt accumulation or real collector geometry.

$$m_{\text{dot_collect}} = LWC * v_{\text{air}} * A * \eta_{\text{collect}} \quad (5)$$

$$Q_{\text{dot_evap}} = m_{\text{dot_evap}} * h_{\text{fg}} \quad (6)$$

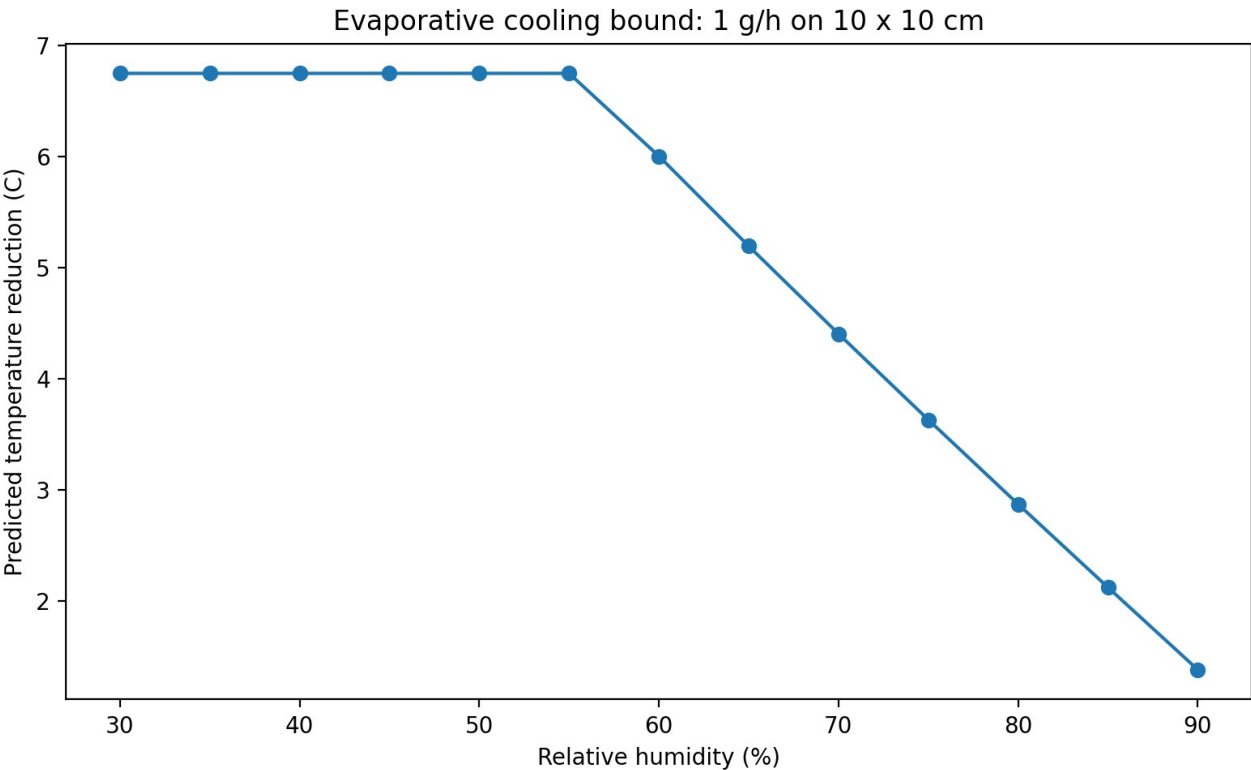


Figure 5. Simple cooling bound for a 10 x 10 cm panel assuming 1 g h⁻¹ evaporation at 30 C. The achievable drop is limited by both latent-heat balance and wet-bulb temperature.

5. Falsification framework

ACS V0.1 was designed around rejection criteria rather than a demonstration-only simulation. The project should be considered unsuccessful in its present form if the central damage-isolation mechanism cannot meet the preregistered hydraulic gates without hidden powered control or unrealistic material behavior.

Table 3. Falsification logic. Passing the computational gates does not substitute for these physical tests.

Hypothesis	Null / failure condition	Pre-registered gate	Physical measurement
H1 Damage-triggered isolation	A puncture cannot recruit enough fluid/mechanical response to choke the leak	>=80% reduction from peak leak within 600 s	Gravimetric or flow-sensor leak rate before/after puncture
H2 Graceful degradation	Isolation destroys useful transport through the surrounding network	>=70% baseline outlet flow retained	Outlet mass flow before and after puncture
H3 Self-sensing	Damage-induced hydraulic state is indistinguishable from noise/drift	Effect >=5 sigma vs matched undamaged control in model; experimental threshold to be preregistered from blank noise	Differential voltage/current with sham-puncture and conductivity controls

Hypothesis	Null / failure condition	Pre-registered gate	Physical measurement
H4 Passive humidity actuation	Gate movement is too weak, too slow or irreproducible	Monotonic aperture change and reproducible hysteresis over cycles	Image-based aperture vs RH, response time
H5 Cooling utility	Wet surface provides no repeatable thermal benefit under controlled airflow	Statistically detectable temperature/heat-flux difference vs dry geometry control	Thermocouples/IR plus mass loss and environmental logging
H6 Water capture	Collector geometry does not exceed blank/control drainage	Positive net capture above blank at matched fog/dew conditions	Mass collected per area per time

6. Simulation design

The nominal network contains 25 nodes arranged as a 5 x 5 grid. Each orthogonal edge receives a radius drawn around a nominal 90 μm mean (180 μm diameter). Edge length is 20 mm and fluid viscosity is 1 mPa s. Left-boundary pressure is set to 1000 Pa and the right boundary to 0 Pa. Damage is introduced at the central node at $t = 300$ s. The nominal collar time constant is 90 s, initial leak conductance is $8 \times 10^{-12} \text{ m}^3 \text{ Pa}^{-1} \text{ s}^{-1}$ and residual conductance is 2% of the initial value.

Table 4. Principal model parameters and sampled ranges.

Parameter	Nominal value	Broad Monte Carlo	Refined design envelope	Interpretation
Mean channel radius	90 μm	65-115 μm clipped distribution	75-105 μm clipped distribution	Fabrication and wetting variability
Source pressure	1000 Pa	700-1300 Pa	800-1200 Pa	Effective capillary / hydrostatic driving scale
Collar time constant τ	90 s	lognormal, median ~120 s	lognormal, median ~90 s; clipped 45-160 s	Speed of isolation response
Residual leak conductance ratio	0.02	broad beta-derived 0.002-0.25	approximately 0.003-0.05	Required degree of closure
Zeta potential	-30 mV	variable around -30 mV	narrower variation around -30 mV	Idealized sensing sensitivity
Conductivity	0.01 S m^{-1}	0.005-0.020 S m^{-1}	0.007-0.014 S m^{-1}	Idealized electrokinetic sensitivity

7. Computational results

In the nominal virtual puncture test, late leak flow was reduced by 92.8% relative to peak leak, while 97.4% of useful outlet flow was retained. The simulation reached an 80% leak-reduction threshold 276 s after puncture. The idealized sensing model showed a large perturbation relative to the assumed 2 μV

noise; because that noise level is an input assumption, the resulting virtual SNR must not be read as measured sensor performance.

7.1 Broad-prior failure is informative

A broad Monte Carlo stress test sampled 1,000 combinations of channel radius, source pressure, hydrogel time constant, residual leak ratio, zeta potential and conductivity. Only 27.3% met all four gates. The dominant lesson is that passive isolation is not guaranteed by redundancy alone: slow swelling or a large residual opening leaves substantial leakage. This failure rate is methodologically important because it prevents the design from being judged only at hand-picked nominal parameters.

7.2 Falsification-driven design envelope

The failed broad-prior trials were converted into a build specification rather than discarded. The refined envelope constrains the collar time constant approximately to 45-160 s and the late residual leak-conductance ratio to 0.003-0.05, while also narrowing geometric and pressure variation. Under those assumptions, 997 of 1,000 Monte Carlo trials passed all computational gates (99.7%). The result is conditional: it says that if a physical collar and network can actually occupy this envelope, the low-order hydraulic model predicts robust isolation. The experiment must now determine whether that premise is true.

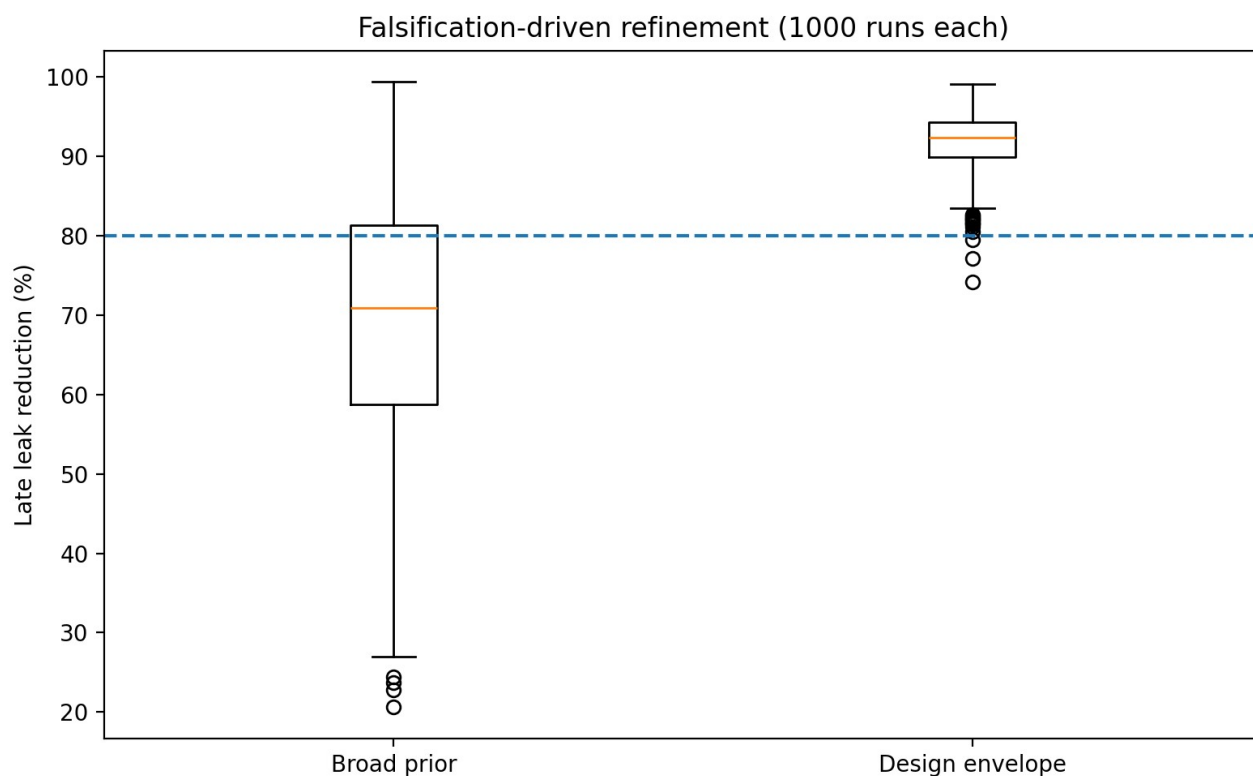


Figure 6. Broad-prior versus refined design-envelope Monte Carlo distributions. The improvement is not empirical optimization; it expresses the material/property envelope required by the model.

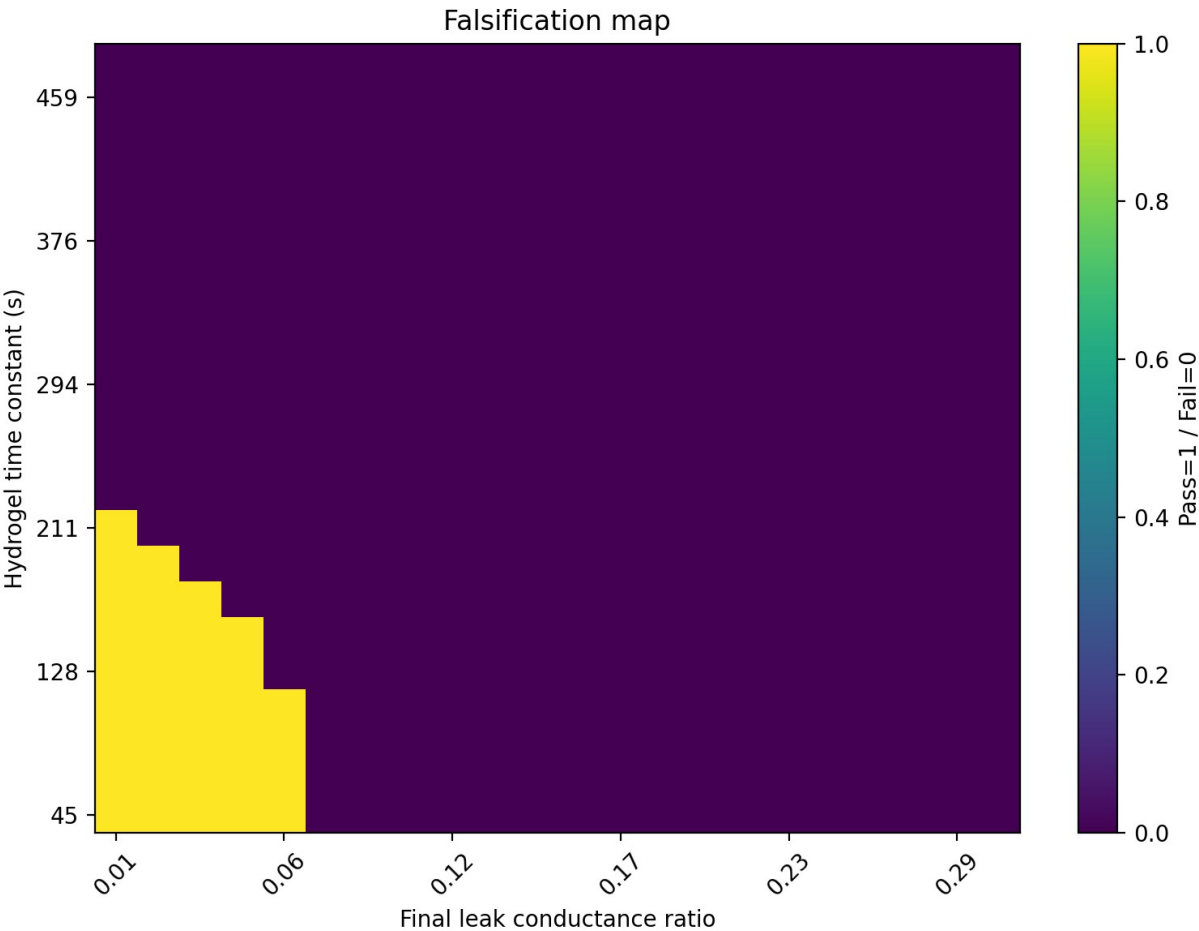


Figure 7. Falsification map over hydrogel response time and residual leak-conductance ratio. Failed regions are as important as passed regions because they define the first fabrication targets.

Table 5. Key computational outputs. None are physical measurements.

Virtual result	Value	Interpretation
Nominal late leak reduction	92.8%	Model prediction only
Nominal useful-flow retention	97.4%	Model prediction only
Nominal time to 80% isolation	276 s	Model prediction only
Broad-prior pass rate	27.3% (273/1000)	Stress-test success rate under broad assumptions
Refined-envelope pass rate	99.7% (997/1000)	Conditional success rate after constraining design properties

8. Physical prototype specification

The first physical build should be deliberately simple: a transparent 100 x 100 mm hydraulic demonstrator that tests the central feedback logic before atmospheric-water or thermal modules are added. A modular stack prevents one failed function from obscuring another.

Table 6. Proposed modular build plan.

Item	Proposed first-build choice	Purpose / measurement	Critical caveat
Transparent vascular plate	Laser-cut, micromilled or molded PMMA/COC/PDMS assembly with channels around 0.18-0.25 mm diameter-equivalent	Visualize priming, routing, bubbles and leak	PDMS swelling/permeability and surface aging may alter wetting
Working fluid	Deionized water first; then controlled electrolyte series	Hydraulic baseline and later electrokinetic tests	Do not mix sensing conclusions across conductivities
Damage port	Replaceable calibrated puncture/orifice cartridge at central node	Reproducible leak conductance	Need geometry calibration rather than arbitrary needle damage
Hydrogel collar	Interchangeable confined PVA/polyacrylate or equivalent swelling ring	Test isolation kinetics and residual aperture	Chemistry must be selected for cycling, leachables and mechanical confinement
Bypass network	At least two alternate routes around central node	Measure graceful degradation	Must separate rerouting from simple upstream flow reduction
Hygromorphic gate coupon	Separate cellulose-based bilayer coupon in V1A	Measure RH response before integration	Should not be integrated until repeatability is known
Electrokinetic node	Charged porous insert with high-impedance electrodes in V1B	Differential self-sensing test	Electrode polarization and ionic drift are major confounders
Capture/cooling surface	Separate top coupon in V1C	Fog/dew collection and evaporation	Requires controlled chamber and environmental normalization

9. Preregistered experimental protocol

1. Calibrate the undamaged network using at least three imposed pressure heads and measure inlet/outlet flow. Reject the hydraulic model if steady flow does not scale with the network solution within a tolerance chosen before fitting.
2. Characterize a calibrated damage orifice without a hydrogel collar to establish peak leak conductance and uncertainty.
3. Install the collar. Repeat puncture trials with randomized specimen order and record leak mass, outlet mass, video, humidity and temperature at ≥ 1 Hz. A sham-damage control receives identical handling without opening the leak.
4. Predefine the primary hydraulic endpoint as percentage reduction in leak from its early post-damage peak at 600 s, with secondary endpoint useful-flow retention. Report every specimen, including failures.
5. For self-sensing, record open-circuit voltage and/or short-circuit current with high-input-impedance instrumentation. Include a geometry-matched uncharged or surface-neutral control, conductivity

series and sham puncture. Do not infer energy harvesting from voltage alone; report source impedance and extractable power.

- 6. For hygromorphic gates, cycle RH in controlled steps and image aperture. Quantify hysteresis, response time, drift and fatigue over a preregistered number of cycles.
- 7. For cooling, compare wet ACS coupon, dry ACS geometry control and a reference surface at matched airflow, radiation and ambient RH. Measure water mass loss so thermal benefit can be reconciled with latent heat.
- 8. For fog/dew capture, report liquid water content or equivalent chamber calibration, airspeed, projected area, collection mass and blank losses.

10. Statistical analysis plan for the physical study

The first physical study is a feasibility experiment rather than a population-level clinical or environmental study. Nevertheless, it should be analyzed with the same discipline as a larger experiment. The primary response variables are continuous: leak reduction, isolation time and flow retention. Raw traces should be retained. Report medians and individual points if distributions are skewed; bootstrap confidence intervals can supplement parametric estimates. A specimen that fails to isolate must not be silently removed as an “outlier”. Any exclusion (for example, fabrication crack discovered before testing) should be defined before looking at the endpoint and documented in the raw-data manifest.

Table 7. Analysis plan for future physical validation.

Comparison	Primary outcome	Recommended analysis
Collar vs no-collar damage	Leak reduction at 600 s	Effect size + confidence interval; paired design if same network can be reused without carryover
Damaged vs sham	Differential electrical signal	Time-aligned effect size; permutation or bootstrap inference if noise is non-Gaussian
Pre- vs post-damage	Useful outlet flow	Ratio with uncertainty; report time course
Wet vs dry surface	Surface temperature / heat flux	Repeated-measures comparison with evaporation mass as covariate/context
Patterned vs blank collector	Collected water flux	Area-normalized difference with chamber replication

11. Raw computational data and reproducibility

The accompanying package contains the full nominal time series, every row from both 1,000-run Monte Carlo ensembles, the complete 22 x 22 falsification parameter grid, humidity-gate predictions, cooling bounds, fog-capture bounds, figures, the spreadsheet view, and the executable model source. A reader can therefore inspect failed runs as easily as successful runs.

Table 8. Data-package manifest.

File	Contents
data/nominal_timeseries.csv	Second-by-second nominal virtual hydraulic and sensing trace
data/monte_carlo_unoptimized.csv	1,000 broad-prior trials including parameters, metrics and pass/fail flags
data/monte_carlo_design_envelope.csv	1,000 refined-envelope trials including parameters, metrics and pass/fail flags
data/falsification_parameter_grid.csv	484-point tau vs residual-leak map
data/hygomorphic_gate_prediction.csv	Opening and closing logistic design curves
data/evaporative_cooling_prediction.csv	Simple wet-bulb / latent-heat cooling bounds
data/fog_collection_bounding_predictions.csv	Fog-collection bounds across liquid water content, airspeed and efficiency assumptions
Adaptive_Capillary_Skin_Simulation_Data.xlsx	Human-readable workbook mirror of the CSV data
acs_simulation_model.py	Reproducible computational core
adaptive_capillary_skin_virtual_prototype.mp4	Conceptual explainer animation; not a finite-element or CFD simulation

12. Limitations and explicit failure modes

- The hydraulic model is steady at each time step and does not resolve meniscus motion, contact-angle hysteresis, bubble nucleation, evaporation inside channels or deformable walls.
- The hydrogel collar is represented by a phenomenological conductance-decay law. Real swelling may be slower, mechanically constrained, anisotropic, irreversible or degraded by salts and repeated cycling.
- The electrokinetic signal is an idealized surrogate. Real electrodes can introduce polarization, redox chemistry, drift and noise; extractable electrical power may be negligible even when open-circuit voltage is measurable.
- The atmospheric-water and cooling calculations are bounding estimates rather than CFD/mass-transfer simulations. Real performance depends strongly on climate, wind, collector geometry, fouling and radiative boundary conditions.
- The individual technologies have substantial prior art. The integrated architecture has not undergone an exhaustive professional patent search, and this document should not be used as a legal opinion.
- Open publication can affect later patent rights in many jurisdictions. Anyone considering patent protection should obtain qualified legal advice before public disclosure.

13. What would make ACS scientifically uninteresting?

A useful falsifiable project needs a stopping rule. ACS should be substantially redesigned or abandoned if controlled experiments show that: (i) damage does not preferentially recruit enough fluid to actuate local isolation; (ii) the collar response cannot achieve the required closure before unacceptable fluid loss; (iii) rerouting cannot preserve useful transport; or (iv) self-sensing adds no information beyond a conventional externally powered pressure/flow sensor at comparable complexity. Likewise, if integration

of the auxiliary functions creates more mass, cost, maintenance or failure points than separate components, the “material metabolism” premise would not provide an engineering advantage.

14. Open-development roadmap

Table 9. Development sequence.

Stage	Goal	Decision gate
V1 - hydraulic cell	Demonstrate puncture -> local swelling -> leak isolation -> rerouting	Proceed only if hydraulic gates are met across independent specimens
V1B - self-sensing	Measure damage-state electrical signature with rigorous controls	Proceed only if signal is repeatable and source impedance/power are characterized
V1C - capture/cooling coupons	Quantify fog/dew collection and evaporative thermal effect independently	Integrate only functions that outperform simple controls
V2 - integrated 10 x 10 cm skin	Couple modules around the same working fluid	Test interference: salts, contamination, evaporation, fatigue
V3 - application prototype	Greenhouse, shelter, façade or environmental patch	Use climate-specific lifecycle/cost analysis before scale-up

15. Data, code and AI-assistance statement

Data availability. All numerical data reported in this version are supplied with the release as CSV files and an XLSX workbook. **Code availability.** The computational core is supplied as `acs_simulation_model.py`. No proprietary experimental data are used.

AI assistance. OpenAI ChatGPT (GPT-5.6 Sol) was used to assist with literature orientation, computational implementation, visualization and drafting. The named human author is responsible for the concept attribution, verification of sources, interpretation, revision, decisions about publication, and any future experimental claims. No AI system is listed as an author.

16. Conclusion

ACS V0.1 is best understood as a disciplined question rather than a finished invention: can the same capillary fluid that a material collects and transports also carry state information and trigger local damage isolation, thereby reducing the need for separate pumps, sensors and controllers? The computational study says this logic is internally possible only inside a constrained response envelope; it does not establish that available materials can occupy that envelope. The next scientifically meaningful step is therefore not another illustration. It is a transparent physical network, a calibrated leak, an interchangeable swelling collar, raw flow traces, matched controls and a willingness to reject the architecture if the effect does not survive measurement.

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Appendix A. Recommended Zenodo record

Suggested resource type: technical note / preprint or dataset-associated publication. Suggested version: 0.1. Suggested keywords: biomimetics; capillary network; hygromorphic materials; hydrovoltaics; atmospheric water; self-healing materials; damage isolation; evaporative cooling; falsification; open hardware. Confirm license, affiliation, ORCID and any IP strategy before upload.

Appendix B. Short reproducibility checklist

- [x] Raw computational CSV files supplied
- [x] Exact Monte Carlo seeds stored as a column
- [x] Pass/fail thresholds stated before physical validation
- [x] Failed virtual trials retained

- [x] Model assumptions and equations stated
- [x] Physical experiments clearly separated from simulations
- [x] No empirical performance claims made from virtual outputs
- [x] Executable source code supplied